

BE IT 2nd YEAR 2nd SEM-2026
OBJECT ORIENTED SYSTEMS
SET-A

Full marks: 30

Time: 1 hour

1. a) Distinguish between each of the following pairs: Provide suitable code snippets for each. [(3x3)=9]
i) final and finally ii) compareToIgnoreCase() and equalsIgnoreCase() iii) Static binding and late binding
2. Fill up the blanks with appropriate phrases. Hence justify the validity of each of the completed sentences with proper arguments. Provide supporting code snippets where necessary. [(2x6)=12]
i) The concept of _____ is achieved in Java by interfaces. It also solves the _____ problem in hybrid inheritance.
ii) Method hiding happens when _____.
iii) We don't have to bother with closing the resources in a program if we _____.
iv) Java is free from the concept of pointers because _____.
v) A class containing atleast one _____ must be declared as _____.
vi) Array of objects is passed to a function by _____. Only one slot of an array of objects is passed by _____.
3. Answer the following questions in brief. Provide suitable code snippets where necessary.
i) Can the two methods be overloaded? Explain with proper reason.
`public int sum(int, int)` and `public void sum(int, int)`
ii) Discuss the advantage of static importing a package with valid code snippet.
iii) If two packages **p1** and **p2** each having a class **A** are imported in a source file, then what problem is created? Show all possible ways of eliminating this problem. [2+2+(2+3)=9]

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Time: 1 hour

[(3x3)=9]

1. Distinguish between each of the following pairs: Provide suitable code snippets for each.
i) ~~throw~~ and ~~throws~~ ii) ~~capacity()~~ and ~~length()~~ methods iii) ~~private~~ and ~~default~~ access specifier

2. Justify the validity (truth/falsity) of each of the following statements with suitable arguments. Provide snippets where needed. [(2x6)=12]

- i) Reference data types do not store the value directly in a variable.
- ii) A private member method in a *Base* class cannot be overridden in its *Child* class.
- iii) All the member methods of a class can be overloaded in a class except the *main()* method.
- iv) Within a static block of a class, we can invoke the non-static member methods of that class.
- v) The equality of two strings can be checked by == operator.
- vi) The *ensureCapacity()* method doubles the capacity of a *StringBuffer* object.

3. Answer the following questions in brief. Provide suitable code snippets where necessary.

i) What do you mean by an anonymous object? Show how it can be created from a class containing one parameterized constructor.

ii) Show how we can invoke several member methods in a class in a cascaded way using anonymous object.

iii) Consider 2 interfaces IN1 and IN2, with IN1 having an abstract *display()* method and IN2 having a *display()* method with its definition. Now create a class *Sample* implementing these 2 interfaces. Show how the *display()* method of the interface IN2 can be invoked using an object of the *Sample* class. Can we invoke the *display()* of IN1? [(1+2)+3+3=9]